

FOLD Progress

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Dataset

- [XNLI](#)
- 15 languages ("ar", "bg", "de", "el", "en", "es", "fr", "hi", "ru", "sw", "th", "tr", "ur", "vi", "zh")
- randomly sampled 600 examples that are at least 20 tokens long for all translations

Baseline

- FSI levels (English to X)
- Mutual intelligibility (Romance, Slavic, Germanic)

Metrics

Overlap

- Likelihood: sum of minimum between A and B, across the frequency bins
- Embeddings: Average of (sum of minimum between A and B, across the frequency bins), across the hidden dimensions

Metrics

KL Divergence

- Likelihood: KL divergence between frequencies of A and B
- Embeddings: Average of (KL divergence between frequencies of A and B), across the hidden dimensions

Metrics

Mean Cosine Similarity

- Likelihood: sum of minimum between A and B, across the frequency bins
- Embeddings: Average cosine similarity between A and B, across the frequency bins

Likelihood, straight spectra `compute_overlaps`

corr_type	var1	var2	coef	pval	w/o coef	w/o pval
pearson	fold	mut_int	-0.547	0.453	-0.547	0.453
spearman	fold	mut_int	0.000	1.000	0.000	1.000
pearson	fsi	fold	-0.061	0.835	-0.088	0.764
spearman	fsi	fold	-0.034	0.908	-0.142	0.629

Likelihood, straight spectra

kl_divergence_matrix

corr_type	var1	var2	coef	pval	w/o coef	w/o pval
pearson	fold	mut_int	0.144	0.856	0.908	0.092
spearman	fold	mut_int	-0.400	0.600	0.800	0.200
pearson	fsi	fold	-0.048	0.871	-0.054	0.856
spearman	fsi	fold	0.440	0.115	0.064	0.829

Likelihood, straight spectra **mae_matrix**

corr_type	var1	var2	coef	pval	w/o coef	w/o pval
pearson	fold	mut_int	0.547	0.453	0.547	0.453
spearman	fold	mut_int	0.000	1.000	0.000	1.000
pearson	fsi	fold	-0.003	0.991	0.059	0.842
spearman	fsi	fold	0.367	0.197	0.401	0.156

Embeddings, straight spectra

`compute_overlaps`

corr_type	var1	var2	coef	pval	w/o coef	w/o pval
pearson	fold	mut_int	0.547	0.453	-0.547	0.453
spearman	fold	mut_int	0.000	1.000	0.000	1.000
pearson	fsi	fold	0.053	0.856	-0.147	0.615
spearman	fsi	fold	-0.186	0.525	-0.293	0.309

Embeddings, straight spectra

kl_divergence_matrix

corr_type	var1	var2	coef	pval	w/o coef	w/o pval
pearson	fold	mut_int	0.568	0.432	NaN	NaN
spearman	fold	mut_int	0.400	0.600	NaN	NaN
pearson	fsi	fold	0.285	0.323	NaN	NaN
spearman	fsi	fold	0.362	0.204	NaN	NaN

Embeddings, straight spectra **mae_matrix**

corr_type	var1	var2	coef	pval	w/o coef	w/o pval
pearson	fold	mut_int	-0.547	0.453	-0.547	0.453
spearman	fold	mut_int	0.000	1.000	0.000	1.000
pearson	fsi	fold	-0.276	0.340	-0.101	0.731
spearman	fsi	fold	-0.337	0.238	-0.352	0.217

Likelihood, welch spectra `compute_overlaps`

corr_type	var1	var2	coef	pval	w/o coef	w/o pval
pearson	fold	mut_int	-0.547	0.453	-0.547	0.453
spearman	fold	mut_int	0.000	1.000	0.000	1.000
pearson	fsi	fold	-0.429	0.126	-0.088	0.764
spearman	fsi	fold	-0.635	0.015	-0.142	0.629

Likelihood, welch spectra

kl_divergence_matrix

corr_type	var1	var2	coef	pval	w/o coef	w/o pval
pearson	fold	mut_int	0.698	0.302	0.908	0.092
spearman	fold	mut_int	0.400	0.600	0.800	0.200
pearson	fsi	fold	0.384	0.175	-0.054	0.856
spearman	fsi	fold	0.743	0.002	0.064	0.829

Likelihood, welch spectra `mae_matrix`

corr_type	var1	var2	coef	pval	w/o coef	w/o pval
pearson	fold	mut_int	0.547	0.453	0.547	0.453
spearman	fold	mut_int	0.000	1.000	0.000	1.000
pearson	fsi	fold	0.429	0.126	0.059	0.842
spearman	fsi	fold	0.635	0.015	0.401	0.156

Embeddings, welch spectra `compute_overlaps`

corr_type	var1	var2	coef	pval	w/o coef	w/o pval
pearson	fold	mut_int	-0.547	0.453	-0.547	0.453
spearman	fold	mut_int	0.000	1.000	0.000	1.000
pearson	fsi	fold	-0.069	0.814	-0.147	0.615
spearman	fsi	fold	-0.362	0.204	-0.293	0.309

Embeddings, welch spectra

kl_divergence_matrix

corr_type	var1	var2	coef	pval	w/o coef	w/o pval
pearson	fold	mut_int	0.584	0.416	NaN	NaN
spearman	fold	mut_int	0.400	0.600	NaN	NaN
pearson	fsi	fold	0.354	0.214	NaN	NaN
spearman	fsi	fold	0.631	0.016	NaN	NaN

Embeddings, welch spectra `mae_matrix`

corr_type	var1	var2	coef	pval	w/o coef	w/o pval
pearson	fold	mut_int	-0.547	0.453	-0.547	0.453
spearman	fold	mut_int	0.000	1.000	0.000	1.000
pearson	fsi	fold	-0.312	0.278	-0.101	0.731
spearman	fsi	fold	-0.606	0.022	-0.352	0.217

Discussion

Mutual Intelligibility Insignificance

- [paper](#) used only calculates *within* 3 language families
- XNLI has only a couple of overlaps

Welch's Method

- [Wikipedia](#)
- Estimates power spectra with a sliding window
- smoothing / noise reduction

Next Steps

Baselines

- find more mutual intelligibility results, perhaps outside language families
- more parallel datasets than XNLI (possibly generate them?)
- find sources for lexical similarity: [ezglot](#), [ethnologue](#)

Next Steps

Coherence / cross spectral density

- [Wikipedia](#)
- the extent to which $y(t)$ may be predicted from $x(t)$ by an optimum linear least squares function
- symmetric
- $0 < c < 1$, 1 means $x(t) == y(t)$
- requires stationarity